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Research Statement

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Research Identity

My research examines how people respond to the design and use of digital platforms and AI systems. I focus on two related questions. First, I examine the psychological mechanisms through which people respond to AI-mediated interactions. Second, I examine the behavioral and economic consequences of platform design using large-scale field data.

I study these questions using two complementary approaches. One stream uses controlled experiments to identify psychological mechanisms such as trust, authenticity, credibility, and cognitive load. The other uses large-scale platform and field data with causal-inference methods to examine behavioral and economic outcomes. Together, these approaches allow me to examine why people respond to digital systems and whether these responses affect behavior at scale. My current project on AI personalization directly connects these approaches by combining a controlled experiment with a field experiment conducted with an e-commerce company.

My current research illustrates this broader agenda. In one study, I examine a loss-based incentive designed to motivate continued effort on a digital platform. I find that experiencing a monetary loss leads to a persistent decline in subsequent performance and increases platform exit. In another study, I examine how AI involvement in product creation affects consumers' perceptions of authenticity. I find that AI involvement affects different dimensions of authenticity depending on the knowledge characteristics of the product, and that these dimensions differ in their effects on purchase intention. Together, these studies examine how the design and use of digital systems influence individual perceptions and behavior.

Human-AI Interaction

My first research stream examines how people respond when AI performs roles traditionally performed by humans. I focus on the psychological mechanisms that shape these responses and how their effects differ across contexts.

My most developed paper in this stream, *"Authenticity in the Age of AI: A Multidimensional Perspective on AI-Generated Products"* (with A. Benedikt Brendel and Alan R. Dennis; target: *MIS Quarterly*), examines how AI involvement in product creation affects consumers' purchase intentions. We distinguish three dimensions of authenticity — constructive, interpersonal, and intrapersonal authenticity — and four knowledge characteristics relevant to product creation: objective, lived-experience, sensory, and imaginative knowledge. In a between-subjects experiment (N = 621), we find that AI involvement affects different dimensions of authenticity depending on the knowledge characteristic most salient to the product. We also find that these authenticity dimensions differ in their effects on purchase intention across products. These findings explain why consumer responses to AI involvement can differ across product categories.

A second project, *"When Multi-Agent Interviews Help or Hurt: Design Principles for Agentic Interview Systems"* (with Lei Wang and Hillol Bala), examines the design of multi-agent AI interview systems. We examine whether agent coordination and role specialization affect the quality of information collected during an interview. We also examine cognitive workload, interviewer credibility, and social presence as mechanisms that may explain these effects. We have developed an IRB-approved experimental platform for two between-subjects studies. Domain experts will evaluate the resulting interview transcripts based on usefulness, actionability, novelty, coverage, and clarity.

A third project, conducted with Hillol Bala, Jane Tan, and Mijalche Santa in partnership with Paket.mk, examines how AI-generated personalized email messages affect customer opens, clicks, conversions, and order amounts. Drawing on Social Identity Theory, we compare surface-level, deep-level, and combined personalization. This project also connects my behavioral and empirical research streams by combining an online experiment, which examines the underlying mechanisms, with a randomized field experiment involving actual customers.

Digital Platforms and User Behavior

My second research stream examines the behavioral and economic consequences of platform design. I use large-scale platform and field data with causal-inference methods to study outcomes such as user performance, retention, and well-being.

My most developed paper in this stream, *“Penalty Scarring: The Hidden Cost of Loss-Incorporating Incentive Designs on Digital Platforms”* (with Seungwook Jin, Keumseok Kang, and Jae Hong Park; currently submitted to *Management Science*), examines incentive designs that impose a loss when users fail to achieve a goal. Prior research has primarily examined whether these designs motivate effort before a loss occurs. We instead examine user behavior after the loss is experienced.

Using data from a digital platform with a deposit-based commitment structure, we exploit variation in the timing of users’ first losses using a staggered difference-in-differences design. We find that achievement rates decrease by approximately seven percentage points following a loss. This decline persists over time and is not reduced when users subsequently choose easier goals. Larger losses are also associated with larger declines in performance. In addition, experiencing a loss approximately doubles the probability of platform exit. We refer to this persistent negative effect as penalty scarring.

A second project, conducted with Rain (Yu) Kan and Lucy (L.) Yan, examines self-disclosure and personality change in the r/ADHD online community. We use a large-scale dataset containing more than 642,000 posts and 6.1 million comments. We extract personality-trait scores using fine-tuned language models and examine whether identity disclosure is associated with subsequent changes in users’ measured Big Five personality traits and continued platform engagement. Our current analysis extends descriptive comparisons using panel fixed-effects regression, event-study designs, and survival models.

The Paket personalization project connects my two research streams. An online experiment examines the mechanisms through which surface-level, deep-level, and combined AI-generated personalization affect consumer response, drawing on Social Identity Theory. A randomized field experiment with Paket.mk customers then examines whether these effects extend to actual opens, clicks, conversions, and order amounts. This design allows us to examine psychological mechanisms in a controlled setting and their behavioral consequences in a field setting.

Methodological Identity

My research combines experimental and empirical methods. I use online and lab experiments to identify psychological mechanisms underlying responses to AI and digital systems. I also use causal-inference methods with large-scale platform and transaction data, including staggered difference-in-differences, panel fixed-effects regression, event-study designs, and survival analysis.

I also use computational text methods when the research question requires measures that are difficult to obtain at scale. These methods include LLM-based content generation and classification and fine-tuned language models for personality-trait extraction. I select methods based on the research question and the type of evidence needed to address it. Controlled experiments allow me to identify psychological mechanisms, while platform and field data allow me to examine whether these mechanisms are reflected in actual behavior.

Future Research Agenda

My future research will extend my current work in three directions.

First, I plan to examine AI-mediated evaluation in additional contexts. My current work examines AI as a product creator and interviewer. I plan to extend this research to settings such as performance evaluation, admissions, and content moderation. This research will examine whether mechanisms such as authenticity and credibility explain responses to AI involvement across different evaluative contexts.

Second, I plan to further connect controlled experiments with field evidence. The Paket project provides one model for this approach by testing a psychological mechanism in controlled experiments and examining its behavioral consequences in a field experiment. I plan to apply similar designs to other questions involving AI and digital platforms.

Third, I plan to extend research that identifies unintended consequences of platform design toward testing alternative designs. My Penalty Scarring project identifies persistent performance and retention costs following loss activation. Future research can examine whether alternative incentive designs maintain motivation while reducing these negative consequences. This direction extends my research from identifying the effects of existing digital systems to evaluating alternative designs.

Together, these research streams examine how digital platforms and AI systems shape individual perceptions and behavior. By combining controlled experiments with causal analysis of field data, my goal is to explain both the mechanisms underlying these responses and their consequences in real settings.